

STAT6061/STAT5008 – Causal Inference

Part 4-3. Instrumental Variable

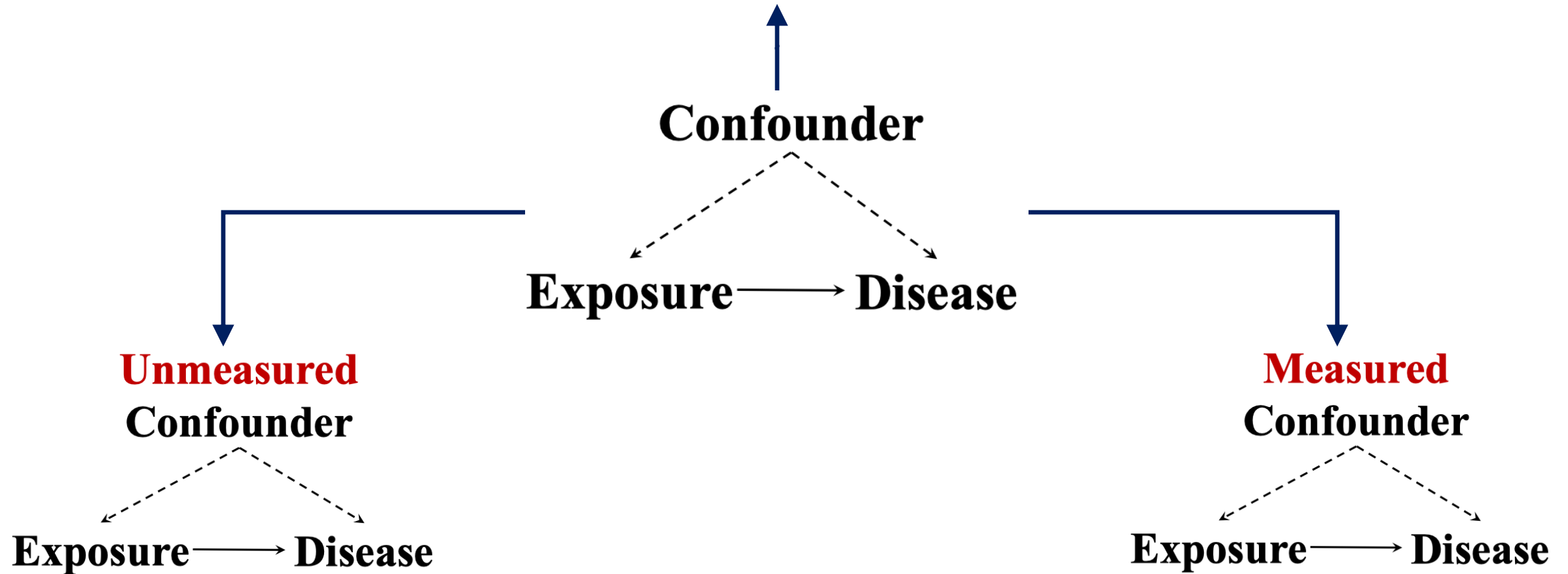
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- *Randomized Experiments*

Exposure —→ **Disease**



- *Front-door criterion*
- *DiD method*
- *Instrumental Variable (Mendelian Randomization)*

- *Standardization (g-formula)*
- *IPW*
- *Matching*

Instrumental variable

(Angrist, Imbens, and Rubin ,1996)

Definition 4.1 (Instrumental Variable).

Let A denote the treatment (or exposure), Y the outcome, and Z an observed variable. We say that Z is a valid *instrumental variable* for estimating the causal effect of A on Y if the following **core IV assumptions** hold:

1. **Relevance:** $Cov(Z, A) \neq 0$

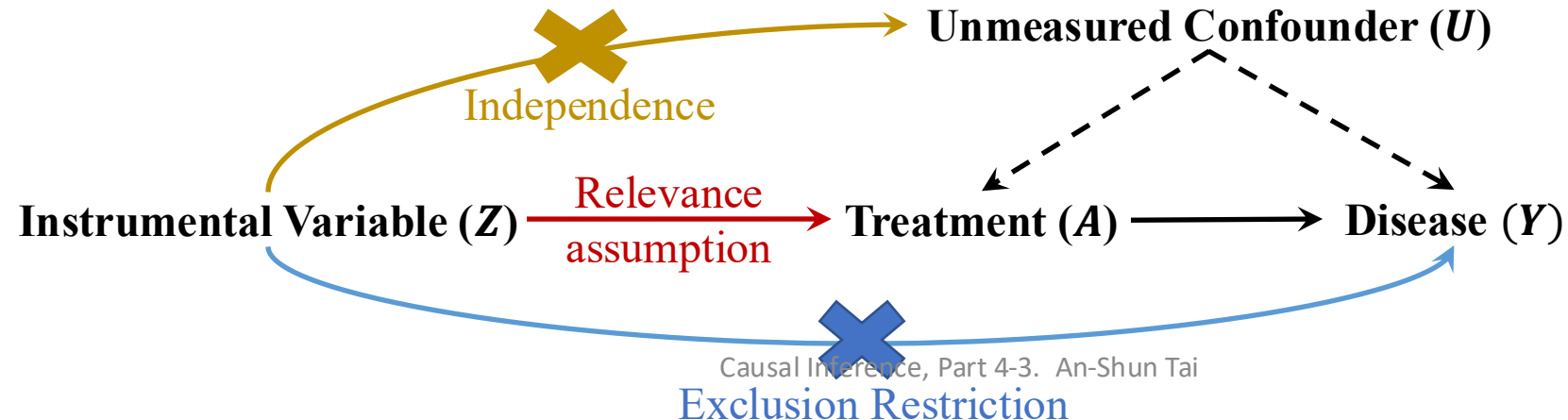
i.e., the instrument Z is associated with the treatment A .

2. **Exclusion Restriction (ER):** $Y(a, z) = Y(a, z') = Y(a)$ for all z and z'

i.e., the instrument Z affects the outcome Y only through the treatment A .

3. **Independence (or Unconfoundedness):** $Z \perp U$,

i.e., the instrument Z is independent of unmeasured confounders U that affect both A and Y .



Intuitive explanation of the three core IV assumptions

- Under the three core assumptions, we can think of Z as a **natural experiment**:
 - It induces exogenous variation in A that is independent of confounding and does not affect Y through other channels.
 - The observed effect of Z on Y reflects the portion of the effect of A on Y that is not contaminated by confounding.
- Example: High-Level neonatal intensive care unit (NICU) and in-hospital mortality (Lorch *et al.*, 2004)
 - Goal: Compare mortality for premature babies in high-level vs low-level NICUs
(High-level = mechanical ventilation & ≥ 50 premature infants/year)
 - Confounding effect: Sicker babies $\rightarrow$ more likely to be in high-level NICUs & more likely to die.
(Unmeasured confounders: fetal heart rate, severity of maternal conditions)
 - IV: Excess travel time ($Z=1$: extra travel time to high-level NICU ≤ 10 min; $Z=0$: extra travel time > 10 min)
Relevance: closer hospitals more likely chosen
Independence: assumes geography unrelated to unobserved severity
Exclusion Restriction: travel time does not directly impact mortality

Sources of instruments in health studies (Baiocchi, Cheng, and Small, 2014)

1. Randomized Encouragement

Example: Encouragement to quit smoking during pregnancy

Key idea: Randomly assigned encouragement (not the treatment itself) serves as the IV

2. Distance-Based Instruments

Example: Distance to nearest specialty hospital or high-level NICU

Key idea: Geographic proximity influences treatment choice but is assumed unrelated to unmeasured confounders

3. Preference-Based Instruments

Example: Physician, hospital, or regional variation in treatment styles

Key idea: Different providers have different tendencies (e.g., some prescribe more aggressively), which creates variation in treatment.

4. Time-Based Instruments

Example: Calendar time, policy implementation dates, or shifts in clinical practice

Key idea: Temporal changes affect treatment assignment but not potential outcomes directly

5. Genetic Instruments (Mendelian Randomization)

Example: Genetic variants (SNPs) influencing cholesterol levels or BMI

Key idea: Genes randomly assorted at conception serve as natural experiments for exposures

6. Administrative or Policy Variables

Example: Insurance plan features (e.g., copay levels, drug formularies)

Key idea: These affect access or adherence to treatment but are assumed not to directly affect the outcome

7. Admission Timing

Example: Time or day of hospital admission (weekday vs. weekend)

Key idea: Treatment protocols may vary by time, creating exogenous variation among otherwise similar patients

Partial identification of the ATE

(Swanson *et al.* 2018)

- The core IV assumptions define the conditions under which **an instrument is considered valid**, but they are not sufficient to identify the ATE from empirical data. Additional assumptions are required to **point-identify** the ATE. (Why? $(Z \rightarrow Y) - (Z \rightarrow A) \stackrel{?}{\Rightarrow} (A \rightarrow Y)$)
- Without additional assumptions, only partial identification of the ATE is possible, meaning the ATE can be bounded within an interval rather than point-identified.

Theorem 4.1 (Manski's worst-case bounds)

Suppose that the outcome and treatment are binary. Under minimal assumptions (e.g., no assumptions about the treatment selection mechanism), the ATE (τ) can be bounded by the observed data as follows:

$$L_{\tau} \leq \tau = \mathbb{E}(Y(1)) - \mathbb{E}(Y(0)) \leq U_{\tau}$$

where

$$U_{\tau} = \Pr(Y = 1|A = 1) \Pr(A = 1) + \Pr(A = 0) - \Pr(Y = 1|A = 0) \Pr(A = 0)$$

$$L_{\tau} = \Pr(Y = 1|A = 1) \Pr(A = 1) - [\Pr(Y = 1|A = 0) \Pr(A = 0) + \Pr(A = 1)]$$

- This bounds always have width 1 and thus include zero. $\Rightarrow$ It cannot determine the direction of the ATE.

Partial identification of the ATE (cont.)

(Swanson *et al.* 2018)

Theorem 4.2 (Robins-Manski bounds, natural bounds)

Suppose that the outcome, treatment, and IV are binary. Under core IV assumptions (e.g., relevance, exclusion restriction, independence), the ATE (τ) can be bounded by the observed data as follows:

$$L_{\tau}^{RM} \leq \tau = \mathbb{E}(Y(1)) - \mathbb{E}(Y(0)) \leq U_{\tau}^{RM}$$

where

$$U_{\tau}^{RM} = \min \left\{ \begin{array}{l} \Pr(Y = 1, A = 1|Z = 0) + \Pr(Y = 0, A = 0|Z = 0) \\ \Pr(Y = 1, A = 1|Z = 1) + \Pr(Y = 0, A = 0|Z = 1) \\ \Pr(Y = 1|Z = 0) - \Pr(Y = 1|Z = 1) + \Pr(Y = 0, A = 0|Z = 0) + \Pr(Y = 1, A = 1|Z = 1) \\ \Pr(Y = 1|Z = 1) - \Pr(Y = 1|Z = 0) + \Pr(Y = 0, A = 0|Z = 1) + \Pr(Y = 1, A = 1|Z = 0) \end{array} \right\}$$
$$L_{\tau}^{RM} = \max \left\{ \begin{array}{l} -\Pr(Y = 0, A = 1|Z = 0) - \Pr(Y = 1, A = 0|Z = 0) \\ -\Pr(Y = 0, A = 1|Z = 1) - \Pr(Y = 1, A = 0|Z = 1) \\ \Pr(Y = 1|Z = 0) - \Pr(Y = 1|Z = 1) - \Pr(Y = 1, A = 0|Z = 0) - \Pr(Y = 0, A = 1|Z = 1) \\ \Pr(Y = 1|Z = 1) - \Pr(Y = 1|Z = 0) - \Pr(Y = 1, A = 0|Z = 1) - \Pr(Y = 0, A = 1|Z = 0) \end{array} \right\}$$

- The width of the natural bounds is no greater than the sum of the noncompliance rates in each arm:

$$\Pr(A = 1|Z = 0) + \Pr(A = 0|Z = 1)$$

A fourth IV assumption

➤ There are two distinct types of fourth IV assumptions used for point identification:

1. Effect Homogeneity (Angrist, 1990):

Assumes that the treatment effect is constant (or mean-independent) across values of the covariates.

$$Y(1) - Y(0) \perp X$$

Under this, the IV estimator identifies the **ATE**.

2. Monotonicity (Imbens & Angrist, 1994; Angrist, Imbens, & Rubin, 1996):

Assumes that the instrument shifts treatment in the same direction for all individuals (i.e., no defiers).

$$A(1) \geq A(0) \text{ for all units}$$

Under this, the IV identifies the **Local Average Treatment Effect (LATE)** for compliers.

A fourth IV assumption: Effect Homogeneity

- The original effect homogeneity assumption (i.e., $Y(1) - Y(0) \perp X$) is the most extreme and historically earliest form of homogeneity.

It is scientifically implausible for most treatments and outcomes and is impossible to satisfy for binary outcomes except under the sharp null or in cases of universal harm or benefit (**Why?**).

- Different versions of effect homogeneity

1. Additive homogeneity across levels of the instrument

$$\mathbb{E}(Y(1) - Y(0)|Z = 1, A = a) = \mathbb{E}(Y(1) - Y(0)|Z = 0, A = a) \text{ for } a = 0, 1$$

- A less extreme assumption than full constant effect.
- However, it is not intuitive, as experts may struggle to justify a constant causal effect across levels Z and A .

2. Effect homogeneity with respect to unmeasured confounders (U)

$$\mathbb{E}(Y(1) - Y(0)|U) = \mathbb{E}(Y(1) - Y(0))$$

- This is a more intuitive form, as it implies that U is not an effect modifier of the treatment effect.
- However, is typically untestable and often implausible.

A fourth IV assumption: Effect Homogeneity (cont.)

- The original effect homogeneity assumption (i.e., $Y(1) - Y(0) \perp Z$) is the most extreme and historically earliest form of homogeneity.

It is scientifically implausible for most treatments and outcomes and is impossible to satisfy for binary outcomes except under the sharp null or in cases of universal harm or benefit (Why?).

- Different versions of effect homogeneity

3. Homogeneity of the instrument–treatment association across levels of U

$$\mathbb{E}(A|Z = 1, U) - \mathbb{E}(A|Z = 0, U) = \mathbb{E}(A|Z = 1) - \mathbb{E}(A|Z = 0)$$

- Testable implications: If some confounders are observed, the instrument–treatment difference should be the same across strata of those confounders.

➤ Remark

- Version 2 and 3 can be founded in Wang and Tchetgen Tchetgen (2018).
- Essentially, these three homogeneity assumptions hold under conditions of no interaction with the relevant variables.
 - Version 1: no $A - Z$ interaction on Y
 - Version 2: no $A - U$ interaction on Y
 - Version 3: no $Z - U$ interaction on A

Identification through effect homogeneity

Consider the following saturated, additive structural mean model for a dichotomous treatment A and an instrument Z

$$\mathbb{E}(Y(1) - Y(0)|A = 1, Z) = \beta_0 + \beta_1 Z$$

- β_0 represents the average treatment effect among treated individuals with $Z = 0$
- $\beta_0 + \beta_1$ is the effect among those with $Z = 1$.
- β_1 captures additive effect modification by Z .

➤ The saturated model can be rewritten as

$$\mathbb{E}(Y - Y(0)|A, Z) = A(\beta_0 + \beta_1 Z)$$

➤ The additive homogeneity across levels of the instrument (Version 1) implies $\beta_1 = 0$ and $\mathbb{E}(Y - Y(0)|A, Z) = A\beta_0$.
 $\Rightarrow \beta_0$ represents the ATT.

➤ Under the ER and independence assumptions, **we have $Y(a) \perp Z$** and
 $\mathbb{E}(Y(0)|Z = 1) = \mathbb{E}(Y(0)|Z = 0)$.

It implies $\mathbb{E}(Y - A\beta_0|Z = 1) = \mathbb{E}(Y - A\beta_0|Z = 0)$.

➤ Solving the above equation, we obtain

$$\beta_0 = \frac{\mathbb{E}(Y|Z = 1) - \mathbb{E}(Y|Z = 0)}{\mathbb{E}(A|Z = 1) - \mathbb{E}(A|Z = 0)}$$

A fourth IV assumption (2): Monotonicity

➤ Principal stratification

Principal strata partition units according to their paired potential treatment statuses under each level of the instrument (IV):

$$(A_i(1), A_i(0))$$

where $A_i(z)$ is the potential outcome of A_i if unit i were assigned $Z = z$.

Stratum	Definition	Notation
Compliers	Take treatment if and only if $Z = 1$	$A_i(1) = 1, A_i(0) = 0$
Always-takers	Always take treatment	$A_i(1) = 1, A_i(0) = 1$
Never-takers	Never take treatment	$A_i(1) = 0, A_i(0) = 0$
Defiers	Do opposite of Z	$A_i(1) = 0, A_i(0) = 1$

➤ Why it matters

Identifies causal effects within subgroups whose stratum membership doesn't depend on assignment.

A fourth IV assumption (2): Monotonicity (cont.)

- In general, principal strata cannot be identified from observed data alone, as they are defined by joint potential outcomes.

Counterfactual values that are never simultaneously observable for any individual.

		Treatment Assignment	
		$Z = 1$	$Z = 0$
Treatment	$A = 1$	Compliers + Always-takers	Always-takers + Defiers
	$A = 0$	Never-takers + Defiers	Compliers + Never-takers

➤ Partial identification of principal strata

1. One-sided compliance: Units in the control group cannot access treatment

$$A_i(0) = 0 \text{ for all } i$$

This implies no always-takers and no defiers.

2. Monotonicity

$$A_i(1) \geq A_i(0) \text{ for all } i$$

This rules out defiers.

A fourth IV assumption (2): Monotonicity (cont.)

➤ One-sided compliance

		Treatment Assignment	
		$Z = 1$	$Z = 0$
Treatment	$A = 1$	Compliers	--
	$A = 0$	Never-takers	Compliers + Never-takers

Examples:

- In a vaccine encouragement trial, only the treatment group receives the invitation; the control group has no access.

➤ Monotonicity

		Treatment Assignment	
		$Z = 1$	$Z = 0$
Treatment	$A = 1$	Compliers + Always-takers	Always-takers
	$A = 0$	Never-takers	Compliers + Never-takers

Examples:

- Rational agents always weakly prefer receiving treatment when assigned to it.
- No "rebellious" behavior (e.g., taking the treatment only when assigned to control).

Identification through monotonicity

➤ Intention-to-treat (ITT) analysis

- It aims to estimate the causal effect of assignment to treatment, regardless of whether participants comply, preserving the benefits of randomization and reflecting real-world effectiveness.

$$ITT = \mathbb{E}(Y(z = 1)) - \mathbb{E}(Y(z = 0))$$

Note: $ATE = \mathbb{E}(Y(a = 1)) - \mathbb{E}(Y(a = 0))$

- The independence assumption (third IV assumption) promises that ITT can be identified as

$$\mathbb{E}(Y|Z = 1) - \mathbb{E}(Y|Z = 0)$$

➤ ITT decomposition

$$\begin{aligned} ITT &= \mathbb{E}(Y(z = 1) - Y(z = 0)|\text{Compliers}) \Pr(\text{Compliers}) \\ &\quad + \mathbb{E}(Y(z = 1) - Y(z = 0)|\text{Always-takers}) \Pr(\text{Always-takers}) \\ &\quad + \mathbb{E}(Y(z = 1) - Y(z = 0)|\text{Never-takers}) \Pr(\text{Never-takers}) \\ &\quad + \mathbb{E}(Y(z = 1) - Y(z = 0)|\text{Defiers}) \Pr(\text{Defiers}) \end{aligned}$$

Step 1: Given the ER assumption,

$$\begin{aligned} \mathbb{E}(Y(z = 1) - Y(z = 0)) &= \mathbb{E}(Y(z = 1, A(z = 1)) - Y(z = 0, A(z = 0))) \\ &= \mathbb{E}(Y(A(z = 1)) - Y(A(z = 0))) \end{aligned}$$

Thus, $\mathbb{E}(Y(z = 1) - Y(z = 0)|\text{Always-takers}) = \mathbb{E}(Y(z = 1) - Y(z = 0)|\text{Never-takers}) = 0$

Identification through monotonicity (cont.)

- Under the ER assumption, ITT decomposition becomes

$$\text{ITT} = \mathbb{E}(Y(z = 1) - Y(z = 0) | \text{Compliers}) \Pr(\text{Compliers}) + \mathbb{E}(Y(z = 1) - Y(z = 0) | \text{Defiers}) \Pr(\text{Defiers})$$

Step 2: We further introduces the **Monotonicity** assumption to rule out the defiers and obtain

$$\begin{aligned} \text{ITT} &= \mathbb{E}(Y(z = 1) - Y(z = 0) | \text{Compliers}) \Pr(\text{Compliers}) \\ &= \mathbb{E}(Y(a = 1) - Y(a = 0) | \text{Compliers}) \Pr(\text{Compliers}) \end{aligned}$$

Step 3: Identification of $\Pr(\text{Compliers})$:

- Consider the causal effect of Z on A

$$\mathbb{E}(A(z = 1) - A(z = 0))$$

- Decomposition

$$\begin{aligned} \mathbb{E}(A(z = 1) - A(z = 0)) &= \mathbb{E}(A(z = 1) - A(z = 0) | \text{Compliers}) \Pr(\text{Compliers}) \\ &\quad + \mathbb{E}(A(z = 1) - A(z = 0) | \text{Always-takers}) \Pr(\text{Always-takers}) \\ &\quad + \mathbb{E}(A(z = 1) - A(z = 0) | \text{Never-takers}) \Pr(\text{Never-takers}) \\ &\quad + \mathbb{E}(A(z = 1) - A(z = 0) | \text{Defiers}) \Pr(\text{Defiers}) \\ &= \mathbb{E}(A(z = 1) - A(z = 0) | \text{Compliers}) \Pr(\text{Compliers}) \\ &\quad + \mathbb{E}(A(z = 1) - A(z = 0) | \text{Defiers}) \Pr(\text{Defiers}) \end{aligned}$$

Identification through monotonicity (cont.)

Step 3: Identification of $\Pr(\text{Compliers})$ (*cont.*):

- Assume **Monotonicity** and we have

$$\mathbb{E}(A(z = 1) - A(z = 0)) = \mathbb{E}(A(z = 1) - A(z = 0) | \text{Compliers}) \Pr(\text{Compliers}) = \Pr(\text{Compliers})$$

- Furthermore, under the **independence** assumption, $\mathbb{E}(A(z = 1) - A(z = 0))$ can be identified as

$$\mathbb{E}(A | Z = 1) - \mathbb{E}(A | Z = 0)$$

As a result, $\Pr(\text{Compliers})$ is identified as

$$\mathbb{E}(A | Z = 1) - \mathbb{E}(A | Z = 0)$$

Step 4: By Steps 1 to 3, we have

$$\mathbb{E}(Y(a = 1) - Y(a = 0) | \text{Compliers}) = \frac{\text{ITT}}{\mathbb{E}(A | Z = 1) - \mathbb{E}(A | Z = 0)} = \frac{\mathbb{E}(Y | Z = 1) - \mathbb{E}(Y | Z = 0)}{\mathbb{E}(A | Z = 1) - \mathbb{E}(A | Z = 0)}$$

➤ The causal effect $\mathbb{E}(Y(a = 1) - Y(a = 0) | \text{Compliers})$ is referred to as the Local Average Treatment Effect (LATE).

- Assessment of LATE tells us the effect for individuals whose behavior can be influenced by the instrument.

The Sommer-Zeger vitamin A supplement data

(Sommer and Zeger, 1991)

- This community-based, cluster-randomized trial in Indonesia evaluated the impact of high-dose vitamin A supplementation on infant survival.
 - Cluster-randomized at the village level.
 - **No infant in control villages received supplement.**
- Data description
 - A total of 23 682 infants were enrolled in the trial.
 - $Z_i = 1$ if the infant's village was randomized to receive high-dose vitamin A, and $Z_i = 0$ otherwise.
 - $A_i = 1$ if the infant actually received the supplement and $A_i = 0$ if not (reflecting noncompliance).
 - $Y_i = 1$ if the infant survived the follow-up period and $Y_i = 0$ if the infant died.

Assignment (Z)	Vitamin supplement (A)	Survival outcome (Y)	Number of Units
0	0	0	74
0	0	1	11,514
1	0	0	34
1	0	1	2,385
1	1	0	12
1	1	1	9,663

Analysis of Sommer-Zeger data

➤ ITT analysis

$$\Pr(Y = 1|Z = 1) - \Pr(Y = 1|Z = 0) = \frac{2385 + 9663}{34 + 2385 + 12 + 9663} - \frac{11514}{74 + 11514} \approx 0.258\%$$

➤ As-treated analysis: control received vs. treatment received

$$\Pr(Y = 1|A = 1) - \Pr(Y = 1|A = 0) = \frac{9663}{12 + 9663} - \frac{11514 + 2385}{74 + 11514 + 34 + 2385} \approx 0.647\%$$

➤ Per-protocol analysis: control received within control assigned vs. treatment received within treatment assigned.

$$\Pr(Y = 1|Z = 1, A = 1) - \Pr(Y = 1|Z = 0, A = 0) = \frac{9663}{12 + 9663} - \frac{11514}{74 + 11514} \approx 0.515\%$$

➤ LATE analysis under monotonicity (or ATE analysis under homogeneity)

$$\frac{\Pr(Y = 1|Z = 1) - \Pr(Y = 1|Z = 0)}{\Pr(A = 1|Z = 1) - \Pr(A = 1|Z = 0)} = \frac{\frac{2385 + 9663}{34 + 2385 + 12 + 9663} - \frac{11514}{74 + 11514}}{\frac{12 + 9663}{34 + 2385 + 12 + 9663} - \frac{0}{74 + 11514}} \approx \frac{0.258\%}{80\%} \approx 0.323\%$$

Assignment (Z)	Vitamin supplement (A)	Survival outcome (Y)	Number of Units
0	0	0	74
0	0	1	11,514
1	0	0	34
1	0	1	2,385
1	1	0	12
1	1	1	9,663

Analysis of Sommer-Zeger data (interpretation)

- LATE analysis under monotonicity (or ATE analysis under homogeneity)

$$\frac{\Pr(Y = 1|Z = 1) - \Pr(Y = 1|Z = 0)}{\Pr(A = 1|Z = 1) - \Pr(A = 1|Z = 0)} = \frac{\frac{2385 + 9663}{34 + 2385 + 12 + 9663} - \frac{11514}{74 + 11514}}{\frac{12 + 9663}{34 + 2385 + 12 + 9663} - \frac{0}{74 + 11514}} \approx \frac{0.258\%}{80\%} \approx 0.323\%$$

Interpretation for the denominator of the IV estimator:

$$\Pr(A = 1|Z = 1) - \Pr(A = 1|Z = 0)$$

- **Under homogeneity (causal estimand: ATE or ATT)**

It is a scaling factor to recover ATE or ATT across the full population. The remaining 20% reflects individuals whose treatment status was unaffected by the instrument.

- **Under monotonicity (causal estimand: LATE)**

It represents the proportion of compliers in the population. The remaining 20% (= 1 - Pr(compliers)) are non-compliers.

Assignment (Z)	Vitamin supplement (A)	Survival outcome (Y)	Number of Units
0	0	0	74
0	0	1	11,514
1	0	0	34
1	0	1	2,385
1	1	0	12
1	1	1	9,663

IV estimator

- Recall: if no unmeasured confounding assumption holds, the *ATE* can be estimated by

$$\widehat{\mathbb{E}}(Y|A = 1) - \widehat{\mathbb{E}}(Y|A = 0).$$

(Ignore the measured confounders C)

- Given a binary instrumental variable (Z), the **IV Estimator (Wald estimator)** is

$$\frac{\widehat{\mathbb{E}}(Y|Z = 1) - \widehat{\mathbb{E}}(Y|Z = 0)}{\widehat{\mathbb{E}}(A|Z = 1) - \widehat{\mathbb{E}}(A|Z = 0)}.$$

1. The IV Estimator is used to assess LATE rather than the ATE. LATE is the treatment effect for the subset of the sample that takes the treatment if and only if they were assigned to the treatment.
2. The IV method can provide useful bounds on the average treatment effect for the whole population.

Wald Estimation:

Ratio of coefficients method (Wald Method) for binary IV

- For a binary IV, and under appropriate assumptions, ATE, ATT, and LATE can each be identified by the same statistical estimand, expressed as:

$$\frac{\mathbb{E}(Y|Z = 1) - \mathbb{E}(Y|Z = 0)}{\mathbb{E}(A|Z = 1) - \mathbb{E}(A|Z = 0)}$$

- This estimand can be directly estimated using the Wald (plug-in) estimator (Angrist, Imbens, and Rubin, 1996).
- Other methods:
 1. **Two-Stage Least Squares** when outcomes and treatments are continuous (Wooldridge, 2010)
 2. **G-estimation** of structural mean models under weaker functional assumptions (Vansteelandt and Goetghebeur, 2003).
 3. **Nonparametric or semiparametric IV methods** (Newey and Powell, 2003).
 4. **IPW in IV contexts for marginal structural models** (Hernán and Robins, 2006).

Wald Estimation: general IVs in linear models

- In practice, a valid IV may consist of **multiple variables or be continuous or multilevel**, in which case the simple identification strategy used for binary IVs no longer applies and fails to identify a single statistical estimand.
- In these scenarios, identifying causal effects requires specifying a model for the outcome:
Linear IV model (Ding, 2024) :

$$Y = \alpha_0 + \alpha_A A + \alpha_u U + \varepsilon_Y, \text{ where } \varepsilon \text{ is an error term.}$$

- α_A represents the ATE (**Why?**)
- This model implies homogeneity (**Which version?**).
- Fitted model: $Y = \alpha_0 + \alpha_A A + \varepsilon_Y^*$, where $\varepsilon^* = \alpha_u U + \varepsilon_Y$. (Notice that the corresponding OLS estimator is biased)

- Property: if the instruments Z is exogenous — **i.e., $Z \perp \varepsilon_Y^*$ which is equivalent to satisfying both the exclusion restriction and independence** — then:

$$\alpha_A = \frac{\text{Cov}(Z, Y)}{\text{Cov}(Z, A)}$$

$$(\text{Cov}(Z, Y - \alpha_0 - \alpha_A A) = 0)$$

Wald Estimation: Other causal effects with binary outcomes

(Mullahy, 1997; Palmer et al. 2011)

- Population causal risk ratio (CRR): $\Pr(Y(1) = 1) / \Pr(Y(0) = 1)$

Outcome: *Log-binomial IV regression model*:

$$\log(\Pr(Y = 1|A, U)) = \theta_0^Y + \theta_A^Y A + \theta_u^Y U$$

Treatment: *Linear model*:

$$A = \theta_0^A + \theta_Z^A Z + \theta_u^A U + \varepsilon_A$$

- θ_A^Y represents the population CRR on the log scale.
- These models also imply homogeneity, specifically that **U is not an effect modifier in any of them.**

- Property: Under these model assumptions, the $\log(\text{CRR})$ can be identified using a valid IV as:

$$\frac{\log(\Pr(Y = 1|Z = 1) / \Pr(Y = 1|Z = 0))}{\mathbb{E}(A|Z = 1) - \mathbb{E}(A|Z = 0)} = \frac{\log RR_{Y|Z}}{\theta_Z^A}$$

- Proof

By the law of total expectation, we have

$$\log(\Pr(Y = 1|Z)) = \log \mathbb{E}(\Pr(Y = 1|A, U, Z) | Z) = \log(\mathbb{E}(e^{\theta_0^Y + \theta_A^Y A + \theta_u^Y U} | Z)) = \log(\mathbb{E}(e^{\theta_0^Y + \theta_A^Y (\theta_0^A + \theta_Z^A Z + \theta_u^A U + \varepsilon_A) + \theta_u^Y U} | Z))$$

Since U and ε_A are independent of Z ,

$$\log(\Pr(Y = 1|Z = 1)) - \log(\Pr(Y = 1|Z = 0)) = \theta_A^Y \theta_Z^A = \theta_A^Y (\mathbb{E}(A|Z = 1) - \mathbb{E}(A|Z = 0)).$$

As a result, the $\log(\text{CRR})$ is identified as $[\log(\Pr(Y = 1|Z = 1)) - \log(\Pr(Y = 1|Z = 0))] / [\mathbb{E}(A|Z = 1) - \mathbb{E}(A|Z = 0)]$

Wald Estimation: Other causal effects with binary outcomes (cont.)

- Population causal odds ratio (COR):

$$\Pr(Y(1) = 1) (1 - \Pr(Y(0) = 1)) / \Pr(Y(0) = 1) (1 - \Pr(Y(1) = 1))$$

Outcome: *logistic IV regression model*:

$$\text{logit}(\Pr(Y = 1|A, U)) = \eta_0^Y + \eta_A^Y A + \eta_u^Y U$$

Treatment: *Linear model*:

$$A = \theta_0^A + \theta_Z^A Z + \theta_u^A U + \varepsilon_A$$

- It's worthy noting that η_A^Y does not represents the population COR or $\log(\text{COR})$ without the rare outcome assumption.
- If the outcome is rare, η_A^Y is the approximation of $\log(\text{COR})$.
- Model structures implies the homogeneity assumption.

- Property: Under these model assumptions and rare outcome assumption, the $\log(\text{COR})$ can be approximated using a valid IV as:

$$\frac{\log[\Pr(Y = 1|Z = 1) (1 - \Pr(Y = 1|Z = 0)) / \Pr(Y = 1|Z = 0) (1 - \Pr(Y = 1|Z = 1))]}{\mathbb{E}(A|Z = 1) - \mathbb{E}(A|Z = 0)} = \frac{\log OR_{Y|Z}}{\theta_Z^A}$$

Two Stage Least Square (TSLS or 2SLS) Estimation

➤ While Wald estimation is computationally simple and can be implemented using either parametric or nonparametric methods, it is limited to the case of a single instrumental variable.

➤ TSLS is a computational method for calculating the causal effect in a two-stage process.

- The underlying true models are assumed to be linear:

$$\begin{aligned} Y &= \alpha_0 + \alpha_A A + \alpha_u U + \varepsilon_Y \\ A &= \theta_0^A + \theta_Z^A Z + \theta_u^A U + \varepsilon_A \end{aligned}$$

- Two-stage process

Step 1: Regress A_i on Z_i , $A_i = \pi_0^A + \pi_Z^A Z_i + \varepsilon_{1i}$, and get fitted values $\hat{A}_i$.

Step 2: Regress Y_i on $\hat{A}_i$, $Y_i = \pi_0^Y + \pi_A^Y \hat{A}_i + \varepsilon_{2i}$, and get the estimate of π_A^Y .

$$\hat{\pi}_A^Y$$

➤ Why does TSLS work?

$$\mathbb{E}(Y|Z) = \mathbb{E}(\alpha_0 + \alpha_A A + \alpha_u U|Z) = \alpha_0 + \alpha_A \mathbb{E}(A|Z) + \alpha_u \mathbb{E}(U) = \alpha_0^* + \alpha_A \mathbb{E}(A|Z)$$

where $\alpha_0^* = \alpha_0 + \alpha_u \mathbb{E}(U)$

Wald Method vs. TSLS

Estimating the causal effect via Wald method

```
> data("iris")
> summary(iris)
  Sepal.Length   Sepal.Width   Petal.Length   Petal.Width   Species
Min.   :4.300   Min.   :2.000   Min.   :1.000   Min.   :0.100   setosa   :50
1st Qu.:5.100   1st Qu.:2.800   1st Qu.:1.600   1st Qu.:0.300   versicolor:50
Median :5.800   Median :3.000   Median :4.350   Median :1.300   virginica :50
Mean   :5.843   Mean   :3.057   Mean   :3.758   Mean   :1.199
3rd Qu.:6.400   3rd Qu.:3.300   3rd Qu.:5.100   3rd Qu.:1.800
Max.   :7.900   Max.   :4.400   Max.   :6.900   Max.   :2.500
> y.reg <- lm(Sepal.Length~Petal.Length,data=iris)
> x.reg <- lm(Sepal.Width~Petal.Length,data=iris)
> coef(y.reg)["Petal.Length"]/coef(x.reg)["Petal.Length"]
Petal.Length
-3.865587
```

Estimating the causal effect via TSLS

```
> x.reg <- lm(Sepal.Width~Petal.Length,data=iris)
> iris$Sepal.Width_prediction <- predict(x.reg)
> summary(lm(Sepal.Length~Sepal.Width_prediction,data=iris))

Call:
lm(formula = Sepal.Length ~ Sepal.Width_prediction, data = iris)

Residuals:
    Min       1Q   Median       3Q      Max
-1.24675 -0.29657 -0.01515  0.27676  1.00269

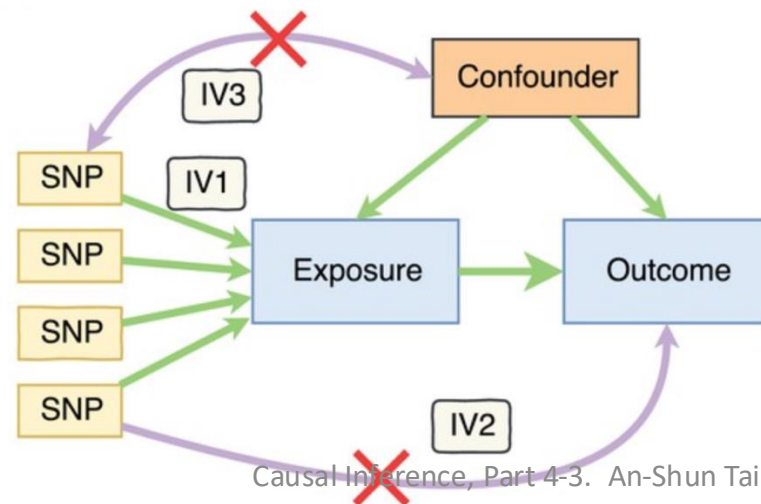
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    17.6617    0.5470   32.29  <2e-16 ***
Sepal.Width_prediction -3.8656    0.1786  -21.65  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4071 on 148 degrees of freedom
Multiple R-squared:  0.76,    Adjusted R-squared:  0.7583
F-statistic: 468.6 on 1 and 148 DF,  p-value: < 2.2e-16
```

The estimates derived from both the Wald method and TSLS are identical!

Why do we need TSLS?

1. TSLS does not inherently restrict the number of instrumental variables
2. In many cases, a single IV (partially for genetic variant) usually accounts for only a small fraction of the variation in a risk factor, often termed as "*weak instruments*," especially in small sample sizes.
3. To address this limitation, researchers have devised approaches utilizing multiple genetic variants that collectively account for a greater portion of the variance in a risk factor compared to a single variant, thereby enhancing statistical power.



Hemani, Gibran, et al. "The MR-Base platform supports systematic causal inference across the human phenome." *elife* 7 (2018): e34408.

Why do we need TSLS? and Why does TSLS work?

Why does TSLS work?

- The slope coefficient obtained from regressing Y on A is

$$\frac{Cov(Y, A)}{Var(A)}$$

Wald Estimation

$$Y_i = \alpha_y + \beta_{y|z}Z_i + \varepsilon_{1i}$$

$$A_i = \alpha_a + \beta_{a|z}Z_i + \varepsilon_{2i}$$

Wald estimator

$$\frac{\hat{\beta}_{y|z}}{\hat{\beta}_{a|z}}$$

TSLS Estimation

$$A_i = \alpha_a + \beta_{a|z}Z_i + \varepsilon_{2i}$$

Step 1: Regressing A_i on Z_i , we have

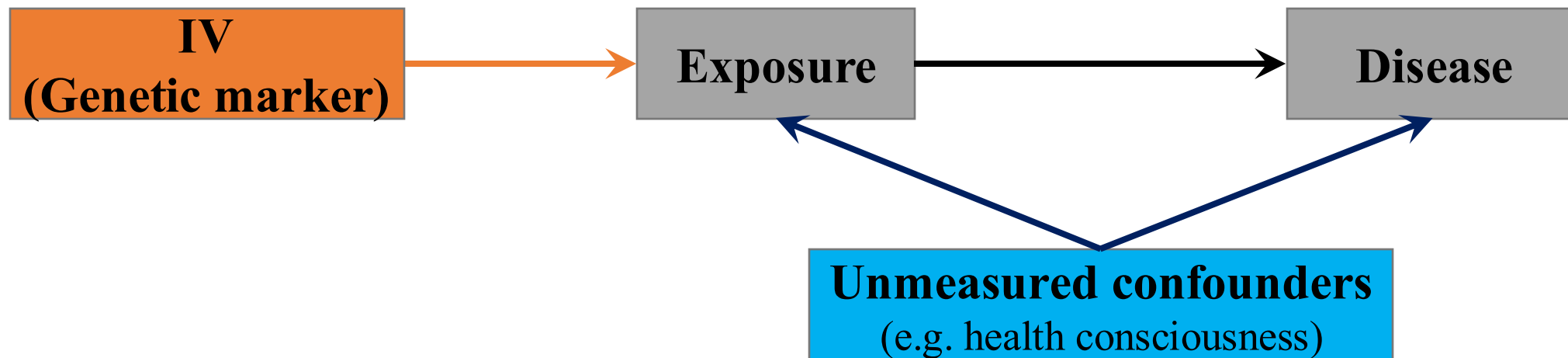
$$\hat{A}_i = \hat{\alpha}_a + \hat{\beta}_{a|z}Z_i$$

Step 2: Regressing Y_i on $\hat{A}_i$, slope coefficient is

$$\begin{aligned}\frac{Cov(Y, \hat{A})}{Var(\hat{A})} &= \frac{Cov(Y, \hat{\alpha}_a + \hat{\beta}_{a|z}Z)}{Var(\hat{\alpha}_a + \hat{\beta}_{a|z}Z)} = \frac{\hat{\beta}_{a|z}Cov(Y, Z)}{\hat{\beta}_{a|z}^2 Var(Z)} = \frac{\hat{\beta}_{a|z}\hat{\beta}_{y|z}Var(Z)}{\hat{\beta}_{a|z}^2 Var(Z)} \\ &= \frac{\hat{\beta}_{y|z}}{\hat{\beta}_{a|z}}\end{aligned}$$

Mendelian Randomization: Mendelian Randomization: A Tool for Causal Inference in Epidemiology

- Mendelian Randomization (MR) uses genetic variants as IVs to estimate the causal effect of an exposure (e.g., BMI) on an outcome (e.g., disease risk).
- Based on Mendel's laws, genetic variants are randomly assigned at conception, mimicking a randomized trial.



Relevance: genetic marker has a causal effect on exposure

Evaluation Focus

- In TSLS MR, focus lies on assessing the F-statistic and R^2 from the first-stage regression of the exposure on the genetic marker (SNP).

F-Statistic: Genetic IV Strength

- The F-statistic reflects the strength of the genetic marker, indicating the size and probability of bias in estimating causal association.
- A "weak" IV (genetic marker) is typically characterized by an F-statistic of less than 10 (rule of thumb).

R^2 : Determining Power

The R^2 from this regression determines statistical power to detect a causal effect, showing the proportion of variability in the exposure phenotype explained by the genetic variant.

Prior Knowledge

Good prior knowledge of variance explained by genetic markers helps determine IV strength and statistical power before study commencement.

Exclusion restriction (ER): genetic marker affects the outcome only through exposure.

Testing the exclusion restriction assumption

- The exclusion restriction assumption cannot be conclusively proven.
- However, statistical, theoretical or biological evidence may bolster ER if such evidence suggests direct pathways are unlikely.

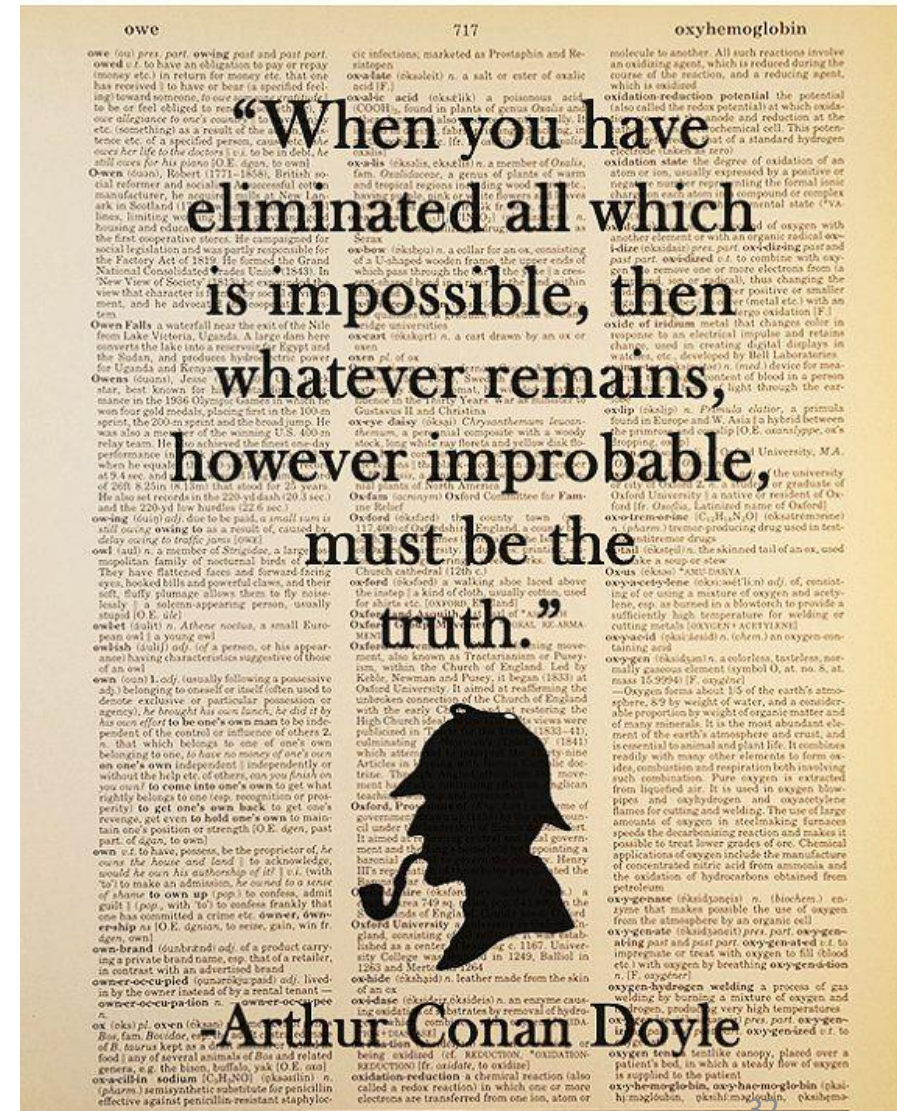
External evidence

External evidence can be used to quantify the genotype-phenotype association; to support the assumption of no other causal pathways between the genotype and the outcome.

Negative control outcomes and Negative control populations

Pleiotropy

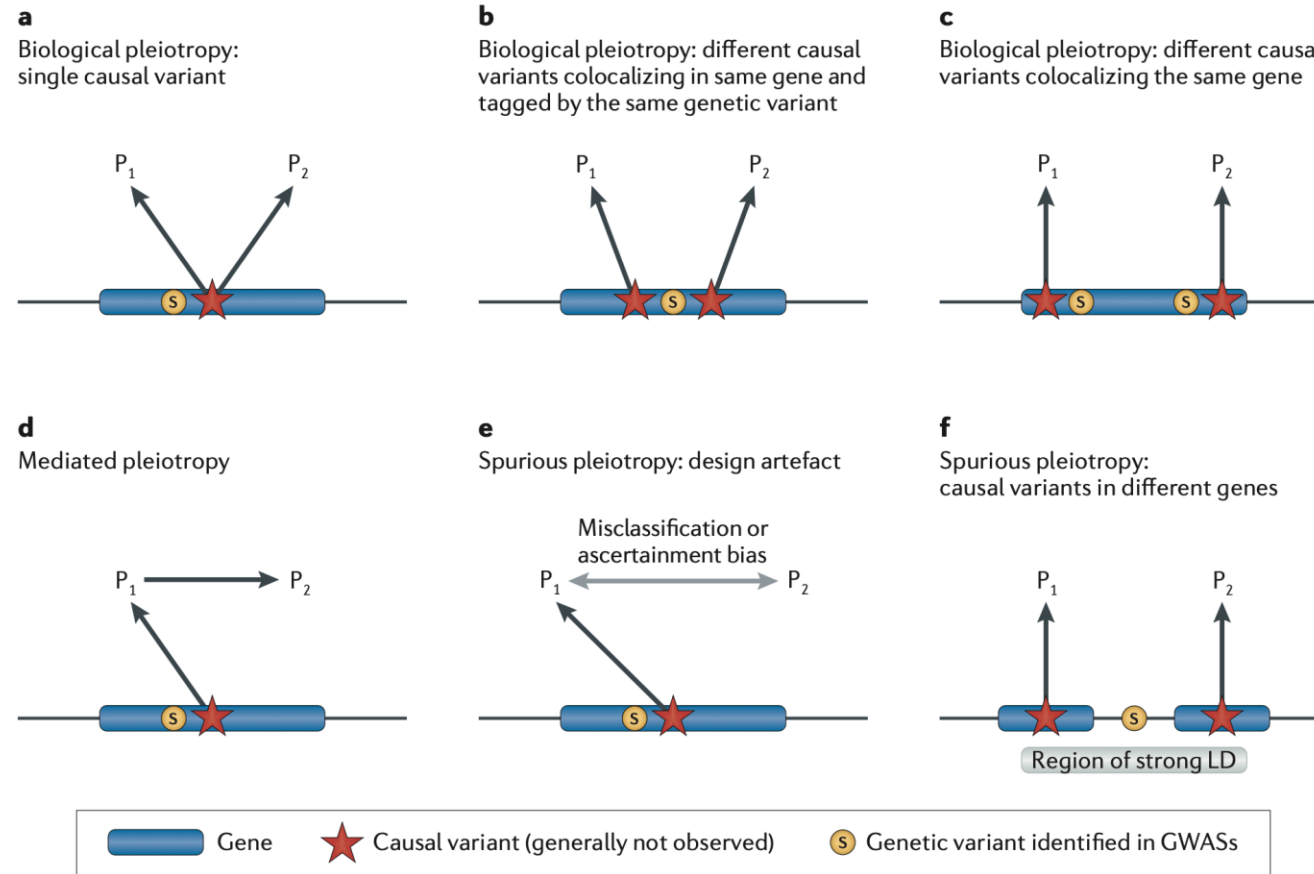
- The phenomenon of a single genetic variant influencing multiple traits



Exclusion restriction (ER): Genetic Pleiotropy

(Solovieff et al., 2013)

- In each scenario, the observed genetic variant (S) is associated with phenotypes 1 and 2 (P_1 and P_2).



Independence: genetic marker does not share common causes with the outcome.

- ✓ Genetic markers, inherited randomly at conception, are independent of environmental factors influencing the outcome.
- ✓ This randomness ensures genetic markers do not share common causes with the outcome, addressing confounding concerns present in observational studies.
- ✓ However, a key concern is **population stratification**, where ancestry may correlate with both the genetic marker and the outcome, potentially confounding MR results.
- ✓ Techniques such as ancestry adjustment or using ancestry-informative genetic markers can help mitigate population stratification effects in MR studies.

Negative control for ER and independence

(Davies et al., 2017)

➤ Negative control outcomes

- Negative control outcomes are likely to be affected by the same confounders as the outcome of interest but are unlikely to be directly affected by the exposure.
- If the genetic marker is associated with a negative control outcome, then this suggests that there may be residual confounding and that exchangeability assumption has been violated.

➤ Negative control population

- A negative control population has a similar confounding structure as the population of interest but was not exposed to the treatment of interest.
- If the proposed genetic marker is associated with any outcomes in the negative control population, this suggests that it may be operating through another mechanism.

One-sample MR and Two-sample MR

IV estimator

$$= \frac{\widehat{\mathbb{E}}(Y|Z = 1) - \widehat{\mathbb{E}}(Y|Z = 0)}{\widehat{\mathbb{E}}(A|Z = 1) - \widehat{\mathbb{E}}(A|Z = 0)}$$

One-Sample MR

✓ Data Structure:

Individual-level data: genetic variants, exposure and outcome measured in the *same* cohort.

✓ Pros:

1. Straightforward implementation in standard IV software.
2. Can adjust flexibly for covariates (e.g. population stratification).
3. Permits direct testing of IV assumptions (e.g. overidentification tests).

✓ Cons:

1. Requires large sample to avoid weak-instrument bias.
2. “Winner’s curse”: SNP–exposure associations estimated in same sample can inflate bias.
3. Measurement error in exposure can bias estimates.

Two-Sample MR

✓ Data Structure:

Summary-level GWAS estimates in two distinct samples.

✓ Pros:

1. Leverages large consortia datasets → greater power.
2. Avoids winner’s curse in SNP–exposure estimates.
3. Permits sensitivity analyses using external summary data.

✓ Cons:

1. Requires non-overlapping samples (or weak overlap) to avoid bias.
2. Limited covariate adjustment; relies on GWAS confounding control.
3. Heterogeneity in study populations may violate comparability.

Multiple instruments: Inverse variance weighted (IVW) method

- The traditional MR method can employ a meta-analysis approach along with the Wald method to estimate the causal effect derived from various genetic markers (SNPs).

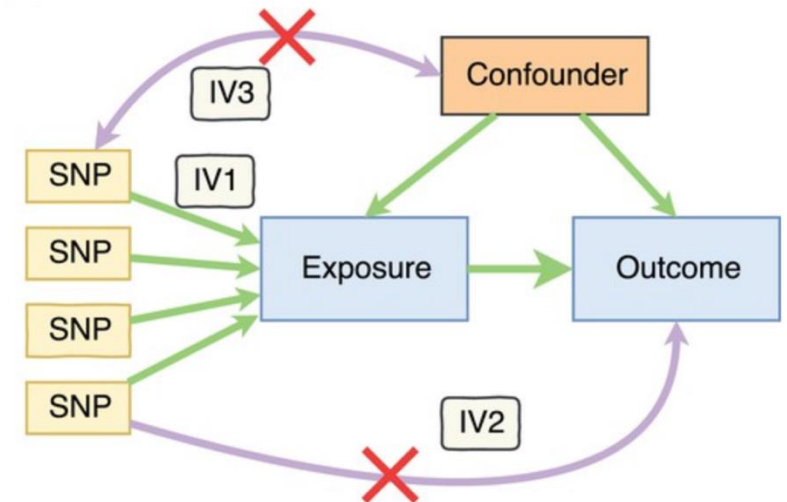
$$Y_i = \alpha_{s,y} + \beta_{s,y|z}Z_{s,i} + \varepsilon_{1i}$$

$$A_i = \alpha_{s,a} + \beta_{s,a|z}Z_{s,i} + \varepsilon_{2i}$$

Wald estimator for SNP- s : $\hat{\tau}_s = \frac{\hat{\beta}_{s,y|z}}{\hat{\beta}_{s,a|z}} = \frac{Cov(Y, Z)}{Cov(A, Z)}$

Wald estimator for pooled data (IVW): $\hat{\tau}_{pooled} = \frac{\sum_s w_s \times \hat{\tau}_s}{\sum_s w_s}$

where $w_s = 1/Var(\hat{\tau}_s)$



Note: IVW estimates are analogous to performing a weighted linear regression of IV-outcome associations against IV-exposure associations, with the constraint that the intercept is zero.

References

Angrist, J. D. (1990). Lifetime earnings and the Vietnam era draft lottery: evidence from social security administrative records. *The american economic review*, 313-336.

Angrist, J. D., Imbens, G. W., & Rubin, D. B. (1996). Identification of causal effects using instrumental variables. *Journal of the American statistical Association*, 91(434), 444-455.

Baiocchi, M., Cheng, J., & Small, D. S. (2014). Instrumental variable methods for causal inference. *Statistics in medicine*, 33(13), 2297-2340.

Davies, N. M., Thomas, K. H., Taylor, A. E., Taylor, G. M., Martin, R. M., Munafò, M. R., & Windmeijer, F. (2017). How to compare instrumental variable and conventional regression analyses using negative controls and bias plots. *International Journal of Epidemiology*, 46(6), 2067-2077.

Hernán, M. A., & Robins, J. M.

(2006). Instruments for causal inference: An epidemiologist's dream? *Epidemiology*, 17(4), 360–372.

References

Imbens, G. W., & Angrist, J. D. (1994). Identification and Estimation of Local Average Treatment Effects. *Econometrica*, 62(2), 467-475.

Lorch, S. A., Baiocchi, M., Ahlberg, C. E., & Small, D. S. (2012). The differential impact of delivery hospital on the outcomes of premature infants. *Pediatrics*, 130(2), 270-278.

Mullahy, J. (1997). Instrumental-variable estimation of count data models: Applications to models of cigarette smoking behavior. *Review of Economics and Statistics*, 79(4), 586-593.

Newey, W. K., & Powell, J. L. (2003). Instrumental variable estimation of nonparametric models. *Econometrica*, 71(5), 1565–1578.

Palmer, T. M., Sterne, J. A., Harbord, R. M., Lawlor, D. A., Sheehan, N. A., Meng, S., ... & Didelez, V. (2011). Instrumental variable estimation of causal risk ratios and causal odds ratios in Mendelian randomization analyses. *American journal of epidemiology*, 173(12), 1392-1403.

References

- Sommer, A., & Zeger, S. L. (1991). On estimating efficacy from clinical trials. *Statistics in medicine*, 10(1), 45-52.
- Swanson, S. A., Hernán, M. A., Miller, M., Robins, J. M., & Richardson, T. S. (2018). Partial identification of the average treatment effect using instrumental variables: review of methods for binary instruments, treatments, and outcomes. *Journal of the American Statistical Association*, 113(522), 933-947.
- Solovieff, N., Cotsapas, C., Lee, P. H., Purcell, S. M., & Smoller, J. W. (2013). Pleiotropy in complex traits: challenges and strategies. *Nature Reviews Genetics*, 14(7), 483-495.
- Vansteelandt, S., & Goetghebeur, E. (2003). Causal inference with generalized structural mean models. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 65(4), 817–835.
- Wang, L., & Tchetgen Tchetgen, E. (2018). Bounded, efficient and multiply robust estimation of average treatment effects using instrumental variables. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 80(3), 531-550.

References

Wooldridge, J. M. (2010). *Econometric analysis of cross section and panel data* (2nd ed.). MIT Press.